

```
pte_t *ptep = pte_offset_map(pmd, va);  
for (; addr < cend; addr += PAGE_SIZE, ptep++) {  
    pte_t pte = *ptep;  
    if (!pte_none(pte)) {  
        if (... && pte_dirty(pte))  
            ...  
    }  
}
```

The diagram consists of two callout boxes with leader lines pointing to specific parts of the code. The first box, labeled "hardwired to radix tree", has a line pointing to the `pte_offset_map` function call. The second box, labeled "In-place read using a pointer to page table entry; no redundant page walks", has a line pointing to the `pte_dirty` function call.

hardwired to radix tree

In-place read using a pointer to page table entry; no redundant page walks